



May University in Cairo

Faculty of Engineering
Robotics, Autonomous and Interactive Systems
Measurement Systems and Sensors Course
2nd Level – 2nd Semester 2025-2026

Date: 2-3-2026

Project Proposal:

Smart Diabetic Foot Ulcer Detection and Prevention Shoe

Design and implement a smart wearable shoe insole that continuously monitors foot conditions using multiple sensors and actively prevents ulcer formation by redistributing pressure and providing corrective feedback.

Team Members:

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| 1. Habiba Tamer Mohamed | ID: 24120470 |
| 2. Yousef Ahmed Mohamed | ID: 24120410 |
| 3. Maryam Ahmed Elsayaed | ID: 24120511 |
| 4. Mohamed Omar Nassar | ID: 24120450 |

Submitted to:

Dr. Shahinaz Hossam
Eng. Mostafa Rashed

Background:

Diabetic foot ulcer is one of the most serious complications affecting diabetic patients and is a major cause of hospitalization and lower-limb amputation worldwide. Foot ulcers typically develop due to prolonged high plantar pressure, poor blood circulation, excessive moisture, and unnoticed inflammation caused by neuropathy. Patients often cannot feel pain or pressure abnormalities, which allows tissue damage to progress without early detection. Continuous clinical monitoring is not always practical outside hospitals, creating a need for wearable systems capable of measuring physiological and biomechanical parameters in real time.

Objectives:

- Design and build a smart insole capable of monitoring multiple ulcer risk factors.
- Measure plantar pressure distribution across the foot in real time.
- Detect localized temperature increase indicating early inflammation.
- Implement an automated motorized swinging mechanism for soothing.
- Monitor humidity levels inside the shoe environment.
- Analyze gait and walking patterns using motion sensing.
- Implement a prevention system that reduces harmful pressure conditions.
- Estimate local blood perfusion using optical sensing.
- Develop a control algorithm that calculates a Foot Ulcer Risk Index.
- Provide alerts and feedback to the user when dangerous conditions occur.
- Ensure low-cost implementation using locally available components.

Scope:

The project will deliver a functional prototype of a Smart Foot Ulcer Detection and Prevention System including:

- Research on diabetic foot pathology and prevention techniques.
- Design of a wearable smart insole structure.
- Integration of pressure, temperature, humidity, motion, and perfusion sensors.
- Development of an active prevention mechanism using vibration feedback.
- Implementation of embedded control logic using a microcontroller.
- Wireless data transmission to a monitoring interface.
- Testing system performance under simulated walking conditions.
- Evaluation of measurement accuracy and response efficiency.
- The system focuses on prevention and early detection rather than medical diagnosis.

Timeframe:

	Description of Work	Start and End Dates
Phase One	Research, medical study, and system design	Week 3-4
Phase Two	Hardware prototyping and sensor integration	Week 6-7
Phase Three	Software development and control algorithms	Week 8-9
Phase Four	Testing, calibration, and final adjustments	Week 10

Project Components:

Mechanical System:

- Specific Shoe design
- Enclosure for electronics

Sensors System:

- Pressure Measurement
- Temperature Monitoring
- Humidity Monitoring
- Motion & Gait Analysis
- Blood Perfusion Monitoring

Control and Electronics:

- Microcontroller (Arduino / ESP 32)
- Analog multiplexer (CD74HC4067) for multiple sensors
- Rechargeable Li-ion battery (3.7V 18650)
- Indicators (Buzzer / LEDs)
- Wires, Connectors and other electronics

References:

- World Health Organization (WHO) – Diabetes Complications Report. (2017)
- International Diabetes Federation – Diabetic Foot Guidelines. (2022)
- IEEE Xplore – Smart Wearable Medical Sensor Systems. (2010)
- Egyptian Knowledge Bank (EKB) – Biomedical Sensor Applications. (2020)
- ScienceDirect – Plantar Pressure Monitoring Systems. (2024)
- IEEE Xplore – IoT-Based Baby Monitoring System (2015)

